ZOO501



OBJECTIVE FOR MID TERM

1: Due to ... of hydrogen, pH rises.

Loss

2: As a result of gastrulation, ectoderm. Mesoderm & ... are formed.

Endoderm

3: Male leopard ... make their sperm in summer.

Frogs

4: ... is a step for gene expression regulation:

Differential gene transcription

5: Coding regions of DNA:

Exons

6: How many domains are present for transcription:

3

7: pdx1 is present in:

Pancreatic And Duodenal Homeobox 1 gene

8: Bacteria transmit genes from one cell to other by:

Sex pili

9: Cells rearrange their ... to create new immunoglobin & antigen receptor gene.

DNA

10: Sea urchin contain ... mRNA.

25,000 to 50,000

11: In many species, sperm are attracted toward eggs by:

Chemotaxis

12: Acrosomal protein mediating recognition by protein ...

Bindin

13: Seawater has ... sodium ion concentration.

High

14: Eggs have sparse, ... spaced yolk & are thus isolecithal.

COLLECTED BY M.KASHIF & MAHAY GULL

Equally

15: Cytoplasm of blastodisc become embryo, this meroblastic division called:

Discoidal

16: The ... constitute major signaling center.

Micromere

17: Osteoblasts secrete collagen-proteoglycan ... that is able to bind calcium salts.

Matrix

18: TGF-β family secreted from the surface ... and roof of neural tube.

Ectoderm

19: Dendrites are numerous and do not extend far from the neuronal cell body or ..., axons may extend for several feet.

Soma

20: One agent of ventral specification is the ... protein.

Sonic hedgehog

21: In frogs & chicks, ... neurulation is seen in neural tube.

Secondary

22: The cells at dorsal most portion of neural tube become:

Neural crest cells

23: About ... of identical twins have two complete & separate chorions.

33%

24: Mammals undergo holoblastic ... cleavage.

Rotational

25: In birds, ... is critical in determining anterior-posterior axis.

Gravity

26: Decidua becomes rich in blood vessels that will provide ... and nutrients to embryo.

Oxygen

27: Mammalian oocyte is released from ovary and swept by fimbriae into the:

Oviduct

COLLECTED BY M KASHIF & MAHAY GULL

28: Fertilization in chick egg occurs in:

Oviduct

29: All multicellular organisms arise by a slow process of progressive change called …

Development

30: Developmental decisions are usually:

Irreversible

31: Major class of cadherin includes:

N-cadherin

32: Only a … of genome is expressed in partial cell.

Small percentage

33: Cortical are … that are membrane bound structure containing proteolytic enzymes.

Granules

34: Small subunit of MPF:

Cyclin-dependent kinase

35: It secretes a … tunic (which gave the name "tunicates" to these creatures).

Cellulose

36: Inner surface cells thin 30 × 5µm process:

Filopodia

37: β-catenin is important for inducing capacity of:

Micromeres

38: In fishes, first … divisions occur synchronously, forming a mound of cells that sits at the animal pole of a large yolk cell.

12

39: Fertilization occurs in … of the oviduct, a region closed to the ovary.

Ampulla

40: In amphibians, such as xenopus, most of the tadpole neural tube is made by …, but the tail neural tube is derived from secondary neurulation.

COLLECTED BY M.KASHIF & MAHAY GULL

Primary neurulation

41: Neural folds migrate toward the midline of the embryo and fusing to form ... beneath the overlying ectoderm.

Neural tube

42: ... form nervous system in the fly or in the frog.

Tissues

43: The marginal zone to form clusters of neurons called:

Nuclei

44: Paracrine factors instruct ... to become muscles by inducing them to synthesize MyoD protein.

Myotome cells

45: Scleraxis is expressed in the mesenchyme ... the sclerotome.

From

46: Acrosomal reaction in ... is often highly specific to each species.

Sea urchins

47: Which type of neurulation is present in fish:

Exclusively secondary

48: Regulation of gene expression by:

Several level

49: ... have two parts of cell cycle.

Blastomeres

50: Number of cortical granules in sea urchin:

15,000

51: How many strategies involve in cell commitment?

2

52: Resting membrane potential of the egg is:

70 mv

53: One chemotactic molecule, a 14-amino acid peptide called:

COLLECTED BY M.KASHIF & MAHAY GULL

Resact

54: Force for sperm propulsion is provided by:

Dynein

55: Regulative development arises from:

Conditional specification

56: Human brain contains ... glial cells.

10^{12}

57: Fishes have discoidal ... pattern.

Telolecithal

58: Process of osteogenesis:

2

59: The mitotic spindle & contractile ring are ... to each other.

Perpendicular

60: Gastrulation in snails is accomplished primarily by:

Epiboly

61: The first cell movement of fish gastrulation is the ... of the blastoderm.

Epiboly

62: This fetal organ the ... is derived primarily from embryonic trophoblast cells.

Chorion

63: Secondary neurulation involves in the making of a ... and its subsequent hollowing into neural tube.

Medullary cord

64: The mantle zone cells differentiate into both ... and ....

Neurons & glia

65: The cerebral cortex in humans has ... layers.

Six

66: Paracrine factors such as ... and ... are important in normal skin development.

TGF & FGF7

COLLECTED BY M.KASHIF & MAHAY GULL

67: **The ... phase involves the invasion of the cartilage model by blood vessels.**

Fifth

68: **Mode of commitment:**

3

69: ***Rana pipiens*** **usually lays around ... eggs while the bullfrog,** ***Rana catesbiana*** **can lay as many as ...**

2500, 20,000

70: **Multinucleate type of cell form:**

Syncytiotrophoblast

71: **Slow block, initiated by the intracellular release of:**

Calcium ions

72: **There are ... major modes of bone formation or osteogenesis.**

2

73: **Mutations of the ... causes camptomelic dysplasis.**

SOX9

74: **Telolecithal is cleavage found in eggs of:**

Birds & fishes

75: **... plays a major role in eye formation.**

Pax6

76: **... and scleraxis are the transcription factors are thought to activate cartilage-specific genes.**

Pax1

77: **The eggs of insects have their yolk in the ... (i.e. they are centrolecithal).**

Center

78: **Egg activates due to which ion more:**

Ca

79: **Which dermis give rise to brain:**

Ectoderm

80: **Which dermis give rise to eye:**

COLLECTED BY M KASHIF & MAHAY GULL

Ectoderm

81: Mode of commitment in insects:

Syncytial specification

82: Nucleosome is composed of an octamer of histone protein wrapped with two loops containing approximately ... of DNA.

140 bp

83: The membrane potential of the sperm becomes ... as potassium ions leave the sperm.

More negative

84: In mouse ZP3 is attached to:

Zona pellucida

85: Fertilin protein is important for sperm-egg fusion in:

Mammals

86: Fast block to polyspermy is initiated by:

Sodium

87: Transition from fertilization to cleavage is caused by factor:

MPF

88: Archenteron is used for:

Primitive gut

89: Spiral cleavage found in:

Snails

90: Sonic hedgehog is a type of:

Protein

91: Neural crest divides functionally in ... domains.

4

92: Myogenic bHLH proteins are specific for the development of:

Muscle cell

93: ... divides mitotically to produce all the cells of the body:

Zygote

COLLECTED BY M.KASHIF & MAHAY GULL

94: ... is the study of the birth defect.

Teratology

95: Identical twins occur in roughly ... of human births.

0.25%

96: Regulative development is seen in most vertebrate embryos and it is obviously critical in development of:

Identical twins

97: Insects such as ... use all three mods of specification to commit their cells to particulate fates.

Drosophila

98: The speed of metamorphosis is directly linked with environmental ...

Pressure

99: In Acetabularia unicellular organisms, development is controlled at both the:

Transcriptional & translational level

100: The individuals of each species, however are divided into two mating types.

Plus & minus

101: Vitelline envelope contains at least ... different glycoproteins.

Eight

102: The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by:

Lazzaro Spallazani

103: The volume of the sea urchin is about ... pico liters.

200

104: Hindbrain:

Rhombencephalon

105: The most crucial, difference b/w mammalian cleavage and all other types involves the phenomenon of:

Compaction

COLLECTED BY M.KASHIF & MAHAY GULL

106: ... is the characteristics of several animal groups, including annelid worms, some flatworm and some mollusks.

Spiral holoblastic cleavage

107: ... animals are those who have ectoderm and endoderm but not true mesoderm, i.e. hydra.

Diploblastic

108: Even distribution of yolk in ovum:

Isolecithal

109: Size of bindin protein:

30,500-Da

110: After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the:

Germ ring

111: Which forms spinal cord, hind brain and mid brain:

Chordamesoderm

112: This streak is first visible in birds as a thickening of the ... at the posterior region of the embryo, just anterior to Koller's sickle.

Epiblast

113: It will induce the uterine cells of the mammals to form the maternal portion of the placenta, the:

Decidua

114: B/w the area pellucida and the area opaca is a thin layer of cells called the ... (or marginal belt).

Marginal zone

115: The mitotic division of the nucleus is called:

Karyokinesis

116: Fishes, ... and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.

Reptiles

117: The ... initiated by sodium influx into the cell in sea urchin.

COLLECTED BY M.KASHIF & MAHAY GULL

Fast block

118: The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the:

Mesodermal somites

119: The rudiment of the central nervous system is called:

Neurulation

120: Blastula produced by the radial cleavage has no blastocoel and are called:

Stereoblastulae

121: Fraternal twins are the result of ... separate fertilization events.

Two

122: The ... induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain and spinal cord.

Prechordal plate

123: The resulting larvae of snail is called ... which resemble tiny bear traps.

Glochidia

124: Fertilization in the absence of sex occur through the:

Binary fission

125: The macromeres divide meridionally, forming a tier of ... cells.

Eight

126: Cells, derived from the micromeres are called the:

Primary mesenchyme

127: A mammalian blastula is called a:

Blastocyst

128: The egg jelly factors that initiate the acrosomal reaction in ... are often highly species specific.

Sea urchins

129: The metencephalon gives rise to the:

Cerebellum

130: In primary neurulation, the original ectoderm is divided into ... sets of cells.

COLLECTED BY M.KASHIF & MAHAY GULL

3

131: The growing region at the tip of the lip is called the:

Chordoneural hinge

132: Neural tube will form:

Brain and spinal cord

133: Each ICM (Inner Cell Mass) cell has the ... potency.

Same

134: Variable cleavages produce no ... fate assignments to cells.

Invariant

135: The neural crest cells originate at the ... most region of the neural tube.

Dorsal

136: The human β-globin gene has an enhancer in its 3′ UTR, roughly ... base pairs.

700

137: The SOX9 gene, ... a DNA binding protein.

Encodes

138: The entire region of calcified spicules becomes surrounded by ... mesenchymal cells that form the periosteum.

Compact

139: The second step is the ... of the myoblasts together into chains.

Alignment

140: Cleavage sphere is known as:

Blastula

141: Osteoclasts are probably derived from the ... precursors as macrophage blood cells.

Same

142: Chabry desired to know the cause of ... defects.

Birth

143: The direct conversion of mesenchymal tissue into ... is called intramembranous ossification.

COLLECTED BY M.KASHIF & MAHAY GULL

Bone

144: Those cells that remain integral components of the neural tube lining become ... cells.

Ependymal

145: The original type of ... cells constitute a layer called the cytotrophoblast.

Trophoblast

146: The major structural characteristics of avian, reptilian & mammalian gastrulation is the:

Primitive streak

147: The blastomere cell cycle is governed by the synthesis and degradation of:

Cyclin

148: Both have anurans:

Toad & frog

149: They have a notochord as larvae (chordates), but they lack:

Vertebrae

150: Acrosomal vesicles is derived from the Golgi apparatus and contains:

Enzymes

151: The larger subunit of MPF:

Cyclin B

152: ZP3 is the specific glycoprotein in:

Mouse

153: MITF is necessary for the production of:

Pigment cells

154: Between the blastoderm and the yolk is a space called:

Sub-germinal cavity

155: The second step of gastrulation involves the:

Involution of the mesoderm

156: The first cleavage furrow appears centrally in the:

Blastodisc

COLLECTED BY M.KASHIF & MAHAY GULL

157: Mother provides fetus with:

Nutrients & oxygen

158: The first segregation of cells within the inner cell mass results in the formation of the:

Hypoblast (sometimes called the primitive endoderm)

159: The fetal portion of placenta formed by:

Chorion

160: Paracrine factors instruct myotome cells to become:

Muscles

161: When myoblasts stop dividing, secrete:

Fibronectin

162: Retinoic acid comes from:

Somites

163: Hox genes pattern the anterior-posterior axis and help to specify positions along that:

Axis

164: The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and ... brain regions.

Hypothalamic

165: Endochondral ossification involves the formation of:

Cartilage

166: J. Hammerling exchanged nuclei b/w two ... distinct species.

Morphologically

167: When plus individuals initiate fusion by extending a fertilization tube, which contacts & fuses with a specific site on the ... individual.

Minus

168: Promoter is ... which is thought to regulate transcription.

Enhancer

169: Identical twins share a common:

COLLECTED BY M.KASHIF & MAHAY GULL

Chorion

170: In birds, ... differences appear crucial for distinguishing dorsal from ventral.

pH

171: The space b/w two neurons, where the transmitters are released:

Synapse

172: Mantle cells differentiate into the:

Gray matter

173: The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area:

Opaca

174: During fertilization, the ... molecules fuse to form the long microfilaments.

Actin

175: A condition in which cell lack the instructions for development is:

Naive

176: Metamorphosis is initiated by hormones from the tadpole's:

Thyroid gland

177: Brain has ... billion neurons.

100

178: Release of Ca from intracellular storage controlled by dye:

Aequorin

179: Allows the mRNA to exit the nucleus:

Poly (A) tail

180: Contact & ... b/w sperm and egg:

Recognition

181: The early embryonic cells need a supply of energy and amino acid which is provided by ... proteins in the egg.

Yolk

182: The depression is called the primitive:

COLLECTED BY M.KASHIF & MAHAY GULL

Groove

183: Red blood cells make:

Globin

184: ... is a discipline that studies embryonic & other developmental processes:

Developmental biology

185: Digestive & respiratory system formed by:

Endoderm

186: Progesterone, work for signals the egg to resume its:

Meiotic division

187: In Chlamydomonas, haploid chromosome has ... copy.

One

188: The calcium ions responsible for cortical granules stores in:

Endoplasmic reticulum of eggs

189: There are ... approaches for anatomy.

4

190: The major cell adhesion molecules appear to be the:

Cadherins

191: Fusion of ... & sperm cell plasma membrane.

Egg

192: The requirement of capacitance varies from species to:

Species

193: The division of embryo into 3:

Germ layers

194: Cell potency ... fate

≥

195: ... also initiates the acrosomal reaction after sperm have bound to it.

Zona Pellucida ZP3

196: The mammalian egg is surrounded by layer called:

COLLECTED BY M.KASHIF & MAHAY GULL

Cumulus

197: The sperm travel by the amoeboid motion of ... of the cell membrane.

Lamellipodial extensions

198: The sex and ... process included into fertilization.

Reproduction

199: The term spermatozoa mean:

Sperm animals

200: In 1824 ... claimed that sperm were not parasites.

J.L. Prevost & J.B. Dumas

201: The second major group of protostomes are the:

Lophotrochozoa

202: Sperms discovered in:

1678

203: In deuterostomes, the mouth opening is formed after:

Anal opening

204: Germ cells formed in specialized portion of:

Egg cytoplasm

205: The third anatomical approach to developmental biology is:

Teratology

206: Fusion is an ... process, often mediated by specific "fusogenic" proteins.

Active

207: Axonemes formed by:

Microtubules

208: Eukaryotic genes are contained within a complex of DNA and protein called:

Chromatin

209: The negative enhancers are called:

Silencers

210: The ability of embryonic cells to change their fates is called:

COLLECTED BY M.KASHIF & MAHAY GULL

Regulation

211: E-cadherin is expressed on all early ... embryonic cells.

Mammalian

212: ... of cell decreases if cell is committed.

Potency

213: Annelids, molluscs and flatworms belong to:

Lophotrochozoan

214: Fertilization in *Rana pipiens* usually lasts for about:

3 years

215: An uncommitted cell can be described as:

Naive

216: Lying immediately beneath the plasma membrane of the eggs is a thin:

Shell

217: Acrosome can be considered as modified secretory:

Vesicle

218: Many marine organisms release their ... into the environment.

Gametes

219: The fast block method is used to prevent:

Polyspermy

220: Autonomous specification was first demonstrated in:

1887

221: TAFs used in:

Transcription factor activity

222: In slow block method which is cortical granule protein:

Hyalin & peroxidase

223: Egg shell formed ... weight of egg.

9 to 12%

224: Potency is ... property.

COLLECTED BY M.KASHIF & MAHAY GULL

Intrinsic

225: A cytoplasm that contains many nuclei:

Syncytium

226: Ectoderm forms:

Epidermis, nervous system, pigment cells

227: According to Dutch sperm are ...

Parasitic animals (spermatozoa)

228: ... constitute the head of the sperm.

Acrosome & nucleus

229: Acrosomal reaction has ... components.

2

230: Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent:

ZP2 molecules

231: Ca & ions may be critical in activating cAMP.

Bicarbonate

232: Peroxidase enzyme ... fertilization envelope.

Hardens

233: The ... is achieved by changing the electric potential.

Fast block to polyspermy

234: P-Cadherin stands for:

Placental cadherin

235: Most transcription factor can bind to:

Specific DNA sequences

236: Outside of the plasma membrane is:

Vitelline envelope

237: Mouse ... is localized to posterior plasma membrane of the sperm head.

Fertilin

COLLECTED BY M.KASHIF & MAHAY GULL

238: Finally, a fourth cortical granules protein ... forms a coating around egg.

Hyaline

239: Enhancers are:

DNA sequences

240: The cleavage stages are called:

Blastomere

241: J. Whittaker strained blastomeres for the presence of the enzyme:

Acetylcholinesterase

242: Protocadherin are involved in separating notochord from mesoderm in:

Xenopus gastrulation

243: The first stand is:

Comparative embryology

244: In protostomes, mouth is formed first than:

Deuterostomes

245: ... can show which cells are transcribing particular genes.

Northern blot, In-situ hybridization & PCR

246: Na & ... are important in fast block to polyspermy.

K

247: Spallanzani demonstrated that filtered ... semen devoid of sperm would not fertilize egg.

Toad

248: Mesoderm generates:

Kidney

249: Mathematical modeling seeks to desire developmental phenomenon in terms of:

Equations

250: A region that is not translated in proteins:

Untranslated region (UTR)

251: RNA polymerase binds with ... in start of transcription.

COLLECTED BY M.KASHIF & MAHAY GULL

Promoter

252: Particular proteins or carbohydrate on the sperm surface ... during capacitation.

Lost

253: Peroxidase enzyme hardens the fertilization envelope by crosslinking ... residues.

Tyrosine

254: ... transmits genes from parents to offspring.

Fertilization

255: Regulation of gene expression accomplished by:

Selective mRNA translation

256: Knowledge of gene activity in humans can be obtained by candidate:

Gene mapping & positional cloning

257: Anatomical abnormalities are caused by:

Mutant genes

258: In several other species, certain regions of the membrane are specialized for sperm recognition & ...

Fusion

259: An ... program has been activated within the cell that causes it to follow a particular pathway of development.

Intrinsic

260: Activation of ... to start development.

Egg metabolism

261: Nuclear control of cell morphogenesis & the interaction of nucleus and cytoplasm are well studied in:

Acetabularia

262: Upon fertilization, the intracellular ... concentration of the egg increases greatly.

Calcium ions

263: Positive and negative mating take place in:

COLLECTED BY M.KASHIF & MAHAY GULL

Chlamydomonas

COLLECTED BY M.KASHIF & MAHAY GULL